Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/26/2016
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Straight Ahead in Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	scott.j.wood@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	(281) 483-6329
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway		
PI Address 2:	Mail code SD2		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	10/01/2011	**End Date:**	10/14/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/14/2016 and new project created due to PI/U.S. CoI change (Ed., 5/10/17)		
Key Personnel Changes/Previous PI:	Drs. Scott Wood and Millard Reschke are U.S. Co-Investigators on this European Space Agency (ESA)-sponsored project, ESA Principal Investigator is Gilles Clément, Ph.D., Lyon Neuroscience Research Center, France.		
COI Name (Institution):	Clement, Gilles Ph.D. (ESA PI: Lyon Neuroscience Research Center) Reschke, Millard (NASA Johnson Space Center)		
Grant/Contract No.:	Not Available		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The subjective straight-ahead direction is a very basic perceptual reference for spatial orientation, movement, and locomotion. The perceived straight-ahead along the horizontal and vertical meridian is largely determined by both otolith and somatosensory inputs. Otolith and somatosensory inputs are altered in microgravity and will change this reference point. Adaptive processes are taking place within the central nervous system to take into account the new environment and compute new spatial egocentric and world-centered representations or frames of reference. This project will measure and monitor how these frames change over time by investigating eye movements and perceptual reports. The three specific aims include: Specific Aim 1: Near & Far Fixation. The first aim is to examine binocular eye movements when subjects fixate on actual targets (normal vision) and then imagine these same targets (occluded vision) in the straight-ahead direction relative to their heading. Initially the subjects' gaze direction and fixation distance will be recorded as they explore the space around them using eye movements in darkness. Next they will be asked to fixate on straight ahead head-fixed targets located at near distance (arm's length, ~0.5 m) and far distance (beyond 2 m). Responses will be compared with different tilt orientations, including pitch tilt forward and backward up to 15 deg. During separate trials, subjects will attempt to maintain fixation on a far Earth-fixed target with and without a vibrotactile sensory aid that indicates how far one has tilted relative to the straight ahead direction. Specific Aim 2: Eye and Arm Movements. The second aim is to examine directed horizontal and vertical eye and arm movements, relative to Earth coordinates and relative to the subject's head/body reference. This task will be performed with the subject upright and then tilted in roll directions up to 30 deg. The trajectory of directed eye and arm movements made in darkness are expected to reflect perceptual tilt errors. Specific Aim 3: Near and Far VOR. The third aim is to examine the influence of target distance on the vestibulo-ocular reflex (VOR) during vertical translation movements. Subjects will stare at actual visual targets (normal vision) at various distances (near and far) in the straight-ahead direction while passively translated up and down using a spring-loaded chair. Vision will then be occluded, and the VOR will be recorded as the subject continues to fixate on the same target locations during translation. In addition to these periodic oscillations (~2.0 Hz), eye movements will also be recorded with vision during unpredictable passive head thrusts up and down using the spring-loaded chair. For each of our specific aims above, our general hypothesis is that responses will be influenced by how accurately subjects perceive their spatial orientation. We will test this hypothesis by comparing responses with and without visual feedback. We also hypothesize for Specific Aim 1 that a vibrotactile sensory aid of tilt position will improve spatial orientation and this reduce gaze fixation errors. Study Participants: Eight International Space Station (ISS) crewmembers will be recruited to participate in three preflight sessions (between 120 and 60 days before launch) and then three postflight sessions on R+0/1 day, R+4 (±2) days, and R+8 (±2) days. Sixteen ground-based subjects will be recruited to participate in a ground control study for up to 3 sessions. A limited number of subjects will also participate in parabolic flight study as resources permit. This study is being implemented by the European Space Agency and is not carried in the U.S. ISS utilization plans. Risk Characterization, Quantification\Evidence: This task will contribute to gap closure by providing information regarding any changes in an individual's egocentric reference that might have negative consequences on evaluating the direction of an approaching object or on the accuracy of reaching movements. This information is important for understanding the problems associated with long-term effects of microgravity on astronauts and how they re-adapt to the return of gravitational forces on Earth or other planetary surfaces. Countermeasure\Prototype Hardware or Software: This task will contribute to gap closure by evaluating how a vibrotactile feedback of reference frames can be used to improve spatial orientation of fixation on space-fixed targets.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	This study will address adaptive changes in spatial orientation as assessed by oculomotor and pointing measures related to the subjective straight ahead, and the use of a vibrotactile sensory aid to reduce perceptual errors. On Earth, there is evidence that patients with vestibular or cerebral lesions present a deviation in their subjective straight-ahead direction. We will test a possible sensory aid countermeasure, vibrotactile stimulation, to improve spatial awareness. This countermeasure may be useful for both astronauts and clinical populations. The results of our flight study also have practical implications in the design of man-machine interfaces. Changes in line of sight in reduced gravity affect crew posture and reach, display orientation, and other visual cues, which should be considered in hardware and operations design.
Task Progress:	Equipment development An existing tilt rotator (Off-Vertical Axis Rotator) was modified to provide tilts in either pitch or roll planes. Subjects are positioned relative to a linear actuator and the tilt orientation can then be computer controlled from upright up to ±30° off-vertical. In addition to the OVAR tilt encoder, body tilt is monitored by a wireless inertial sensor (Xsense) strapped to the torso and wrist. Arm pointing measurements are also obtained by tracking a hand laser projected to an adjacent screen. Eye movements will be recorded with a lightweight binocular video mask (SensoMotoric Instruments Inc., Needham, MA) and the recorded pupils are tracked using a custom eye tracking algorithm implemented on a Linux operating system. Before getting into the chair, subjects will don a belt that has up to eight small electromechanical vibrators (tactors) positioned around the torso to provide vibrotactile feedback of tilt position. Vibrotactile feedback is provided during some trials using commercial tactors (C2 model, Engineering Acoustics Inc.) that operate similar to the vibration mode on cell phones. Each tactor can be cycled from a slow pulse rate to steady on to indicate both direction and magnitude of tilt. Flight study Eight International Space Station (ISS) crewmembers will be recruited to participate in three preflight sessions (between 120 and 60 days before launch) and then three postflight sessions on R+0/1 day, R+4 (±2) days, and R+8 (±2) days (Clément and Wood 2016a). Preflight data was initiated in 2015 following approval for this study to be implemented for pre- and post-flight testing only. To date, one ISS crewmember has completed pre- and post-flight data collection. Preflight data collection has been obtained from four other crewmembers, although one of these subjects was withdrawn from the study due to changes in post-flight test plans. Ground control studies: Parabolic flight: During the definition phase of the study, preliminary data collection was obtained during parabolic flights, first during Novespace flights in Bordeaux, France (2011-2012) and subsequently during ZeroG flights in Houston, TX (spring 2013). This purpose of this parabolic study was to examine the spatial coding of eye movements during roll tilt relative to perceived orientations while free-floating during the microgravity phase of parabolic flight or during head tilt in normal gravity (Specific Aim 2). Binocular videographic recordings obtained in darkness allowed us to quantify the mean deviations in gaze trajectories along both horizontal and vertical coordinates relative to the aircraft and head orientations. Both variability and curvature of gaze trajectories increased during roll tilt compared to the upright position. The saccades were less accurate during parabolic flight compared to measurements obtained in normal gravity. The trajectories of saccades along perceived horizontal orientations tended to deviate in the same direction as the head tilt, while the deviations in gaze trajectories along the perceived vertical orientations deviated in the opposite direction relative to the head tilt. Although subjects were instructed to look off in the distance while performing the eye movements, fixation distance varied with vertical gaze direction independent of whether the saccades were made along perceived aircraft or head orientations. This coupling of horizontal vergence with vertical gaze is in a consistent direction with the vertical slant of the horopter. The increased errors in gaze trajectories along both perceived orientations during microgravity can be attributed to the otolith's role in spatial coding of eye movements (Wood and Clément 2013). Pilot vibrotactile feedback was also implemented in the parabolic flights in Johnson Space Center (JSC). Subsequent to this pilot study, the experiment was redefined as a pre- and post-flight study only due to delays in the flight equipment development. Test-retest repeatability: Following the definition of the pre- and post-flight study protocol, a ground control study was initiated to obtain normative data on 16 healthy non-astronaut subjects participating in 3 sessions similar to the astronaut preflight data sessions. The initial analysis has focused on the linear VOR measures that utilize a spring loaded chair design similar to Field Test Dynamic Visual Acuity (PI M. Reschke). The translational Vestibulo-Ocular Reflex (tVOR) is an important otolith-mediated response to stabilize gaze during natural locomotion. In the initial phase of this study, we began to examine an existing data set obtained during Off-Vertical Axis Rotation. In this data set, we were able to establish that the modulation of horizontal slow phase velocity was larger at higher frequencies (>0.8 Hz) than lower frequencies (<0.05 Hz). Previously described kinematic effects of fixation distance were not present for these lower frequencies (Clément and Wood 2016b). Therefore, for our Straight Ahead in Microgravity (SAM) study protocol, we limited the tVOR trials to only high frequency trials that simulate the kinematic demands of natural locomotion. An additional goal of our control SAM study was to develop a measure of the tVOR using our simple hand-operated chair that provided passive vertical motion. As with the flight study, binocular eye movements were recorded with a tight-fitting video mask. Vertical motion was provided by a modified spring-powered chair (swopper.com) at ~2 Hz (±2 cm displacement) to approximate the head motion during walking. Linear acceleration was measured with wireless inertial sensors (Xsense) mounted on the head and torso. Eye movements were recorded while subjects viewed near (0.5 m) and far (~4 m) targets, and then imagined these targets in darkness. Subjects also provided perceptual estimates of target distances. Based on preliminary data on ten subjects, the tVOR gain was greater with near targets, and greater with vision than in darkness. We concluded that this portable chair system can provide a field measure of otolith-ocular function at frequencies sufficient to elicit a robust tVOR (Wood et al. 2016).
Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Wood SJ, Campbell D, Reschke MF, Prather L, Clément G. "Vestibulo-ocular responses to vertical translation using a hand-operated chair as a field measure of otolith function." Presented at the 39th Annual MidWinter Meeting of the Association for Research in Otolaryngology, San Diego, CA, February 20-24, 2016. Association for Research in Otolaryngology Abstracts. 2016;39:287. , Feb-2016
Abstracts for Journals and Proceedings	Wood SJ, Clément G. "Spatial coding of eye movements relative to perceived orientations during roll tilt with different gravitoinertial loads." Presented at the Neuroscience 2013, San Diego, CA, November 9-13, 2013. Neuroscience 2013, San Diego, CA, November 9-13, 2013. Program#/Poster#: 367.01/DDD9. Available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=3236&sKey=bd78a188-c200-481a-9287-07920d4ad5c3&cKey=3b09fa4b-c286-4fc4-8b57-36806a887641&mKey=8d2a5bec-4825-4cd6-9439-b42bb151d1cf , Nov-2013
Abstracts for Journals and Proceedings	Clément G, Wood SJ. "Subjective straight ahead after adaptation to weightlessness." Presented at the International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales Joint Life Science Meeting 'Life in Space for Life on Earth,' Toulouse, France, June 5-10, 2016. International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales Joint Life Science Meeting 'Life in Space for Life on Earth,' Toulouse, France, June 5-10, 2016. http://esaconferencebureau.com/2016-events/16a13 , Jun-2016
Articles in Peer-reviewed Journals	Clément GR, Wood SJ. "Translational otolith-ocular reflex during off-vertical axis rotation in humans." Neurosci Lett. 2016 Mar 11;616:65-9. Epub 2016 Jan 28. https://dx.doi.org/10.1016/j.neulet.2016.01.049 ; PubMed PMID: 26827718 , Mar-2016